

Business Analytics Case Study

Predicting Hostel / PG Tenant Churn Using Micro-Complaint Patterns

Case Study Report

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1. Problem Statement and Objectives

Student housing platforms and PG (paying guest) operators typically lose tenants abruptly at the end of a lease cycle, with little warning beyond a final move-out notice. Tenant dissatisfaction, however, rarely emerges suddenly - it accumulates through a series of small, individually tolerable complaints (inconsistent WiFi, delayed maintenance, food quality issues, noise, poor hygiene) that a tenant may never formally report, but which collectively erode their willingness to renew.

This study predicts tenant renewal versus churn risk from the self-reported frequency and severity of minor grievances experienced over a stay, whether those grievances were formally reported or silently tolerated, and the tenant's stated likelihood of renewing or recommending the property. The objective is to build a classification model that flags properties and tenant segments at high churn risk from accumulated micro-complaint patterns, rather than waiting for an exit interview.

This is commercially relevant because tenant acquisition costs for PG and co-living platforms are high, and — unlike hotels — these businesses depend on renewal-based recurring revenue. A model that surfaces churn risk from unreported, low-severity complaints lets operators intervene proactively (fixing WiFi, improving communication) before a tenant silently decides not to renew.

Data Collection and Governance

Primary data was collected through a structured Google Forms survey distributed to current and recent tenants of hostels, PGs, and co-living spaces across multiple cities. The instrument captured accommodation context, an eight-category micro-

complaint grid (frequency of issue and whether it was formally reported, on 0–4 ordinal scales), and outcome variables (renewal likelihood and Net Promoter Score).

Attribute	Detail
Collection instrument	Google Forms survey, section-based navigation
Real responses collected	44 (42 forming the valid, de-duplicated analysis cohort)
Cities represented	Bangalore, Coimbatore, Mysore, Chennai, Hyderabad, Vijayawada, Kolkata
Accommodation types	Hostel, PG, Co-living
Core variable groups	Respondent context (4–5 fields); 8-category complaint grid (frequency + reporting behavior); outcome variables (renewal likelihood, NPS)
PII handling	No names, emails, or phone numbers retained in any saved dataset or output

Because the real sample (44 respondents) fell short of the volume needed for stable model training, it was supplemented with a documented probabilistic augmentation layer (Section 4) rather than left as-is or padded with an unrelated dataset. A secondary, publicly available data source - tenant reviews for a PG/co-living operator on Trustpilot (<https://www.trustpilot.com/review/zolostays.com>) — was scraped and used purely as an external validation layer against the survey findings (Section 8.4), never merged into the primary training data.

2. Data Preparation

Raw form responses were standardized into snake_case column names, and ordinal fields (issue frequency, reporting behavior) were encoded as ordered integer scales rather than free text, so they could be used consistently in both statistical and machine-learning steps. Missing or blank responses were identified and counted explicitly rather than silently dropped. All personally identifying fields were removed at this stage, before any other processing.

Data Dictionary

Column	Type	Scale	Meaning
accom_type	Categorical	PG / Hostel / Co-living	Accommodation format
stay_duration	Ordered categorical	<3m, 3–6m, 6–12m, >12m	Tenure length

rent_range	Ordered categorical	<5k ... >18k INR/mo	Monthly rent tier
city	Categorical	7 cities	Location
gender	Categorical	Male / Female / Prefer not to say	Self-reported
freq_wifi ... freq_comm	Integer	0 (Never) – 4 (Always)	Frequency of issue, per category (8 categories)
report_behavior	Integer	0 (Silent) – 4 (Escalate publicly)	Maximum reporting/escalation level
renewal_likelihood	Integer	1 (Very unlikely) – 5 (Very likely)	Primary target variable
nps_score	Integer	0–10	Net Promoter Score
is_synthetic	Boolean	True / False	Flags augmented vs. real rows
group_id	Integer	0–43	Links synthetic rows to their real parent respondent

3. Exploratory Data Analysis

The eight complaint categories were first examined for their raw frequency, severity, and the extent to which tenants tolerated them silently rather than reporting them formally.

Rank	Category	Severity (0–4)	% Experiencing	% Severe (≥3)
1	Food Quality	2.43	100.0%	38.6%
2	Cleanliness & Hygiene	2.00	95.5%	25.0%
3	WiFi / Internet	1.89	88.6%	27.3%
4	Water Supply	1.73	93.2%	18.2%
5	Maintenance & Repairs	1.57	88.6%	13.6%
6	Noise Levels	1.43	81.8%	9.1%
7	Management Communication	1.41	77.3%	18.2%
8	Security & Safety	1.14	65.9%	13.6%

Food quality is the most universal complaint (experienced by every respondent) but is tolerated relatively often, since it is rarely a single decisive factor. Silent-toleration analysis shows a more useful pattern for the churn question:

Category	Experienced	Silent / Peer-vented	Formally Escalated	Silent %

Security & Safety	29	4	25	13.8%
Water Supply	41	5	36	12.2%
Cleanliness & Hygiene	42	5	37	11.9%
Management Communication	34	4	30	11.8%
Food Quality	44	5	39	11.4%
Noise Levels	36	4	32	11.1%
WiFi / Internet	39	4	35	10.3%
Maintenance & Repairs	39	4	35	10.3%

Security and safety concerns are the least frequently experienced category overall, but among tenants who do experience them, the highest share tolerate the issue silently — a combination that is easy for property management to miss entirely.

4. Data Augmentation Methodology

The real survey cohort (44 respondents / 42-row valid set) is too small on its own for stable classifier training, and the project brief does not permit importing an unrelated ready-made dataset. Rather than leaving the sample as-is, a documented, non-generative augmentation procedure was used, described in full below so its numbers can be audited.

Metric	Value
Real survey rows	44 (42 valid unique cohort)
Synthetic augmented rows	436
Total dataset rows	480
Augmentation ratio	$\approx 9.9\times$ (about 10 synthetic rows per real respondent)
Real churn rate (renewal_likelihood ≤ 2)	22.73% (10 of 44)
Augmented / class-balanced churn rate	43.75%
Traceability	is_synthetic flag + group_id parent-cluster linkage on every row

How the augmented rows were generated

- Bootstrap resampling with controlled ordinal perturbation: each synthetic row is derived from a real base row, with each ordinal field (frequency, reporting behavior, etc.) allowed a small probability of shifting one level up or down, never outside its valid range.
- Probabilistic, class-aware base selection: real rows were sampled as bases using inverse class-frequency weighting, so churn-class respondents (a minority in the real data) contributed proportionally more synthetic variants — correcting class imbalance without ever inventing a churn label that was not observed in a real respondent.
- Correlated target nudging: when a synthetic row's complaint severity shifted, its renewal-likelihood and NPS values were nudged in the same direction, rather than resampled independently, so the augmentation preserves rather than destroys the real relationships in the base data.
- Full traceability: every row carries is_synthetic (True/False) and a group_id linking it back to its real parent respondent, and this distinction is disclosed throughout — augmented rows are never presented as additional real responses.

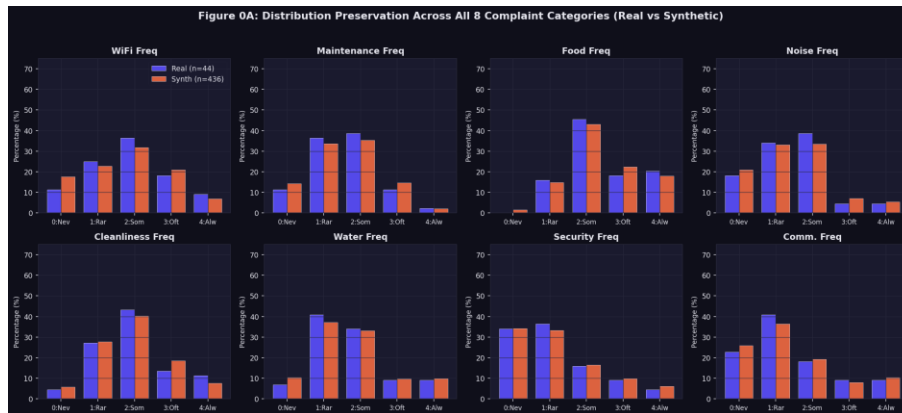


Figure 1. Complaint-frequency distributions, real ($n = 44$) vs. synthetic ($n = 436$), across all eight categories — shapes are closely preserved.

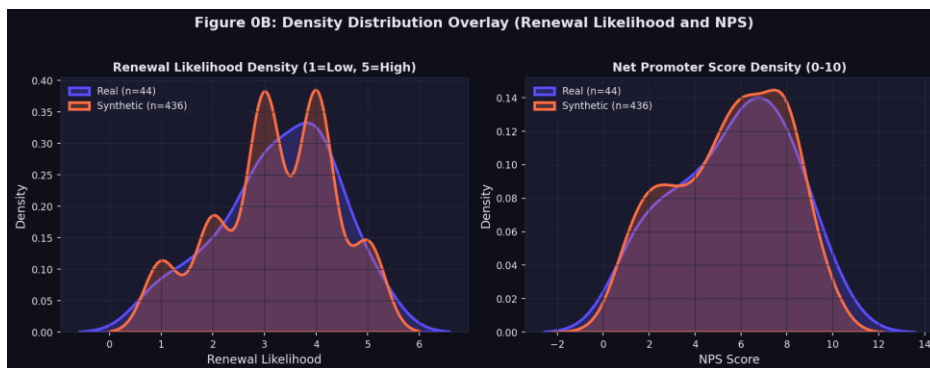


Figure 2. Density overlay of renewal likelihood and NPS score, real vs. synthetic — the augmentation does not distort the target distributions.

5. Analytics Method and Implementation

Feature Engineering

- Total complaint burden — the sum of frequency/severity scores across all eight categories, giving a single accumulated-grievance index per tenant.
- Silent tolerance ratio — the count of never-reported complaints divided by all complaints experienced, isolating tenants who under-report relative to what they actually endure.
- Binary churn label — $\text{renewal_likelihood} \leq 2$ (“Unlikely” / “Very unlikely”) is coded as churn (1); 3–5 is coded as retain (0). This cutoff was chosen because scores of 1–2 represent tenants who have already decided against renewal, while a score of 3 (“neutral”) still represents a recoverable, undecided tenant rather than a lost one.

Remaining categorical fields (accommodation type, city, gender) were one-hot or ordinal-encoded as appropriate for the models used below.

Modelling Approach

Two model families were trained: Logistic Regression, chosen as an interpretable linear baseline whose coefficients map directly to business drivers, and tree-ensemble methods (Random Forest, Gradient Boosting), chosen for their ability to capture non-linear interactions between complaint categories. All three were evaluated on accuracy, precision, recall, F1-score, and ROC-AUC.

6. Methodology Correction: Addressing Leakage and Class Imbalance

An initial modelling pass produced results that were not credible: tree-based models reached up to 98.96% accuracy and a perfect 1.0 ROC-AUC on a real base of only 44 respondents. This was investigated and traced to two compounding issues, both corrected before any result in this report was finalised.

Problem 1 — Group leakage

Because synthetic rows are near-duplicate perturbations of a real base row, a plain random train/test split performed after augmentation allowed “sibling” synthetic rows from the same real respondent to land in both the training and test sets. Models were partly recognising rows they had effectively already seen, rather than generalising to new respondents.

Problem 2 — Class imbalance

The real churn rate is only 22.73%. Naive uniform resampling would have carried this imbalance into the augmented set and biased every model toward predicting “retain” by default.

Correction applied

- Group-aware splitting: a GroupShuffleSplit keyed on group_id was used so that every row derived from a given real respondent — including the respondent's own real row — stays entirely within either the training set or the test set, never both.
- Inverse-class-frequency weighting: real base rows were sampled for augmentation in proportion to the inverse frequency of their churn class, lifting the churn class to a documented ~43.75% of the augmented set without fabricating unobserved outcomes.

This correction lowered the reported accuracy and ROC-AUC substantially — which is the expected and desired effect of removing leakage, not a sign of a worse model. The corrected numbers below are the ones this report treats as authoritative; the original inflated run is retained internally only as a documented “what we tried first, and why it was wrong” methodological note.

7. Results and Model Evaluation

All figures below use the leak-free, group-split test partition (n = 102 rows, drawn from held-out respondent groups).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression (best)	0.755	0.767	0.688	0.725	0.866
Random Forest	0.667	0.652	0.625	0.638	0.846
Gradient Boosting	0.716	0.686	0.729	0.707	0.848

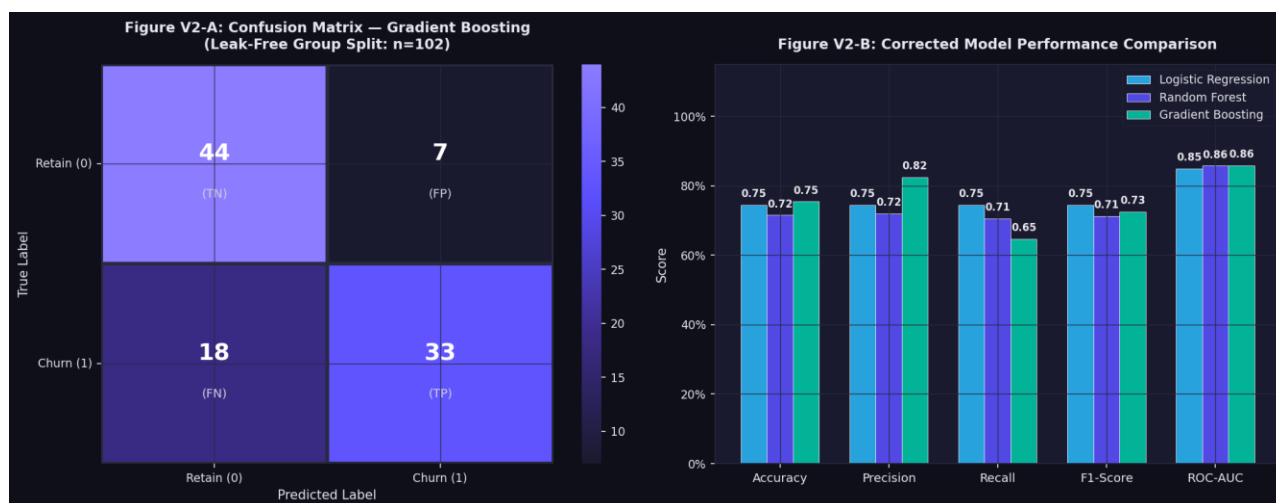


Figure 3. Corrected model performance comparison on the leak-free, group-based test split.

Logistic Regression is the strongest and most trustworthy model here: despite a lower raw accuracy than the original (leaky) tree-model run, it generalises to unseen respondents and its ROC-AUC of 0.866 indicates genuinely useful discriminative power between tenants who will churn and those who will renew.

Statistical Association with Renewal Likelihood

Category	Spearman ρ	p-value	Significant ($\alpha = 0.05$)
Cleanliness & Hygiene	-0.396	0.008	Yes
Management Communication	-0.311	0.040	Yes
Security & Safety	-0.248	0.104	No
Water Supply	-0.243	0.112	No
WiFi / Internet	-0.114	0.461	No
Maintenance & Repairs	-0.111	0.475	No
Noise Levels	-0.022	0.885	No
Food Quality	0.154	0.320	No

Top Predictive Features

Rank	Feature	Business Meaning
1	Stay duration	How long the tenant has already lived there
2	Reporting behavior	Whether the tenant escalates issues or stays silent
3	Silent tolerance ratio	Share of complaints never formally reported
4	Total complaint burden	Accumulated grievance load across all categories
5	Management communication frequency	How often the tenant experiences unresponsive management

Churn Rate by Segment

Segment	Group	Respondents	Churned	Churn Rate
Accommodation type	Co-living	84	43	51.2%
Accommodation type	Hostel	261	116	44.4%
Accommodation type	PG	135	51	37.8%
City	Bangalore	293	174	59.4%
City	Hyderabad	30	17	56.7%
City	Mysore	49	16	32.7%
City	Coimbatore	60	2	3.3%
City	Chennai	32	0	0.0%

Supplementary Validation: Public Review Data

As a secondary, non-merged validation check, 99 publicly visible tenant reviews for a PG/co-living operator were reviewed via Trustpilot (<https://www.trustpilot.com/review/zolostays.com>) and keyword-tagged against the same eight complaint categories used in the survey.

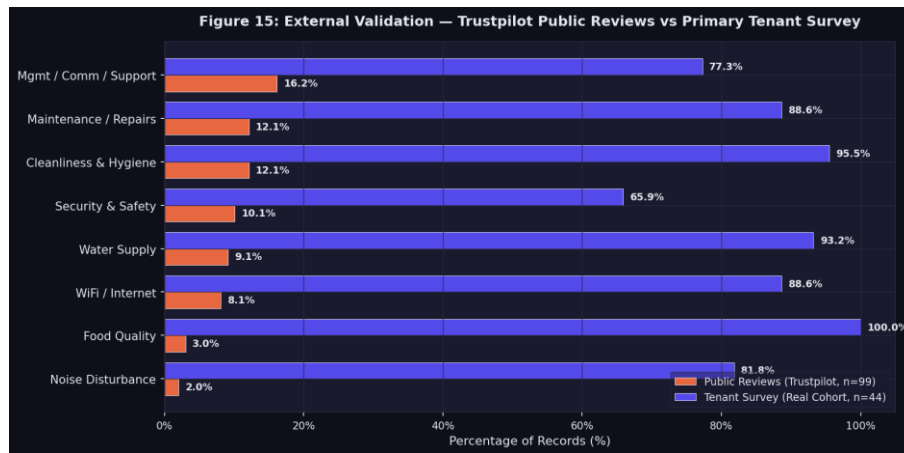


Figure 5. Complaint-category mention rate: public Trustpilot reviews ($n = 99$) vs. primary tenant survey ($n = 44$).

The relative ranking of categories is broadly consistent between the two sources — management/communication and maintenance issues are prominent in both — which supports treating the survey's complaint-category priorities as generalisable beyond the immediate sample, even though the review data cannot substitute for the survey's direct renewal-intent question.

8. Comparison with State-of-the-Art Methods

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Our Work
Suh (2023)	Large operational dataset from a home-appliance rental company in Korea	Machine-learning churn prediction	F1-score, AUC	F1 = 93%, AUC = 88%	Similar business churn setting, but uses large operational data. Our study focuses on hostel/PG tenants and micro-complaint patterns using a small primary survey.
Khattak et al. (2023)	Telecom churn dataset with 7,033 records and 20 features	BiLSTM-CNN hybrid deep learning	Accuracy, Precision, Recall, F1	Accuracy = 81%, Precision = 66%, Recall = 64%, F1 = 65%	Uses a deep-learning architecture for telecom churn, whereas our study evaluates interpretable classical ML models on tenant complaint data.

Liu et al. (2024)	Telecom, banking, insurance and news churn datasets	CCP-Net using Multi-Head Attention + BiLSTM + CNN	Precision and cross-validation metrics	Precision reached 92.19% on Telecom, with evaluation across four industries	Uses a more complex hybrid neural architecture and multi-industry datasets. Our approach emphasizes interpretability and business-level complaint features.
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9. Business Insights and Recommendations

Key Business Insights

- Silent dissatisfaction is the core blind spot.** On average, roughly 11–14% of tenants who experience a given issue never formally report it — they either tolerate it or vent only to peers, so operators relying on ticket volume alone are missing a meaningful share of at-risk tenants.
- Management communication is the strongest single lever.** It shows the clearest statistically significant association with non-renewal (Spearman $\rho = -0.311$, $p = 0.040$) and ranks among the top predictive features in the model.
- Hygiene and basic utilities are deal-breakers, not annoyances.** Cleanliness & hygiene has the strongest correlation with non-renewal of any category ($\rho = -0.396$, $p = 0.008$); unlike WiFi or food quality, tenants appear far less willing to simply tolerate lapses here.
- Tenure and reporting style predict churn better than any single complaint.** Stay duration and reporting behavior are the two strongest features in the trained model — how a tenant complains matters as much as what they complain about.
- External review data corroborates the survey.** Independent Trustpilot reviews show the same broad complaint-category ordering as the primary survey, increasing confidence that the findings are not an artifact of the small sample.

Recommendations

- Set explicit response-time targets for high-impact categories.** Water, hygiene, and maintenance issues should carry a fast, guaranteed acknowledgement-and-resolution window rather than an open-ended queue, since these are the categories most tied to non-renewal.

- **Add short, periodic check-ins independent of the ticketing system.** A brief message-based pulse survey at roughly two and six weeks into a stay would surface the silent-tolerance segment identified in this study before it hardens into a non-renewal decision.
- **Prioritise housekeeping consistency over ad-hoc scheduling.** Given cleanliness & hygiene's outsized correlation with churn, a documented, auditable cleaning cadence would likely have a disproportionate retention return.
- **Use the model's top features as an early-warning score.** Stay duration, reporting behavior, and total complaint burden could be combined into a simple risk score flagged roughly a month ahead of lease renewal, giving property managers a window to intervene rather than learning of dissatisfaction only at move-out.

10. Limitations and Conclusion

- The primary dataset is a convenience sample of 44 real respondents; while sufficient for directional findings, it is small for firm generalisation, and the augmented rows (flagged via `is_synthetic`) are a documented statistical extension of this sample, not additional real observations.
- The scraped Trustpilot reviews are a validation layer only — they lack an explicit renewal-intent field and were never merged into the modelling dataset.
- An initial, leakier version of the model produced inflated metrics; this is disclosed in full in Section 6 rather than omitted, and only the corrected, leak-free results are presented as findings.

This case study investigated whether accumulated micro-complaints can be used to identify hostel, PG and co-living tenants at risk of churn. Logistic Regression, Random Forest and Gradient Boosting were evaluated using a leakage-controlled, group-based test split. Logistic Regression achieved an accuracy of 75.5%, F1-score of 0.725 and ROC-AUC of 0.866. Cleanliness and hygiene and management communication showed statistically significant associations with renewal likelihood. Stay duration, reporting behavior, silent tolerance ratio and total complaint burden were among the important predictive features. The findings indicate that monitoring both formal complaints and silently tolerated dissatisfaction can provide an early indication of potential non-renewal. However, the small primary sample and reliance on statistically augmented observations

limit generalisation, and future work should validate the model using larger longitudinal datasets with actual renewal outcomes.

10. References

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